

Week 5 Assignment

Handling Missing Data • SWID Data Analytics Programme

Due: Before the next class

Background

You have just joined a small e-commerce company as a junior data analyst. The sales team has handed you a dataset of customer orders — but like most real-world data, it is not clean. Your job is to analyse the data, identify the problems, and clean it properly before any analysis can happen.

The Dataset

Copy the code below into your notebook to create your working dataset:

```
import pandas as pd
import numpy as np

data = {
    'Order_ID': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    'Customer_Name': ['Amara', 'Bola', None, 'Diana', 'Efua', 'Fatima', 'Grace',
None, 'Ifeoma', 'Joyce'],
    'Country': ['Ghana', 'Nigeria', 'Kenya', None, 'Ghana', 'Cameroon', None,
'Kenya', 'Nigeria', 'Ghana'],
    'Product': ['Laptop', 'Phone', 'Tablet', 'Laptop', None, 'Phone', 'Tablet',
'Laptop', None, 'Phone'],
    'Quantity': [2, 1, None, 3, 2, None, 1, 4, 2, None],
    'Price': [1500, 800, 600, 1500, None, 800, 600, 1500, None, 800],
    'Delivery_Days': [3, None, 7, 5, 3, None, 6, None, 4, 5]
}

df = pd.DataFrame(data)
print(df)
```

Your Tasks

Task 1 — Explore the data

Before cleaning anything, take a good look at what you are working with. Run the three commands below and write two to three sentences describing what you find. How many missing values are there? Which columns are affected?

```
df.info()
df.describe()
df.isnull().sum()
```

Task 2 — Handle missing customer names

Some orders have no customer name. You cannot recover that information, but you still want to keep the order in the dataset. Fill those missing names with the word **"Unknown"** and write one sentence explaining why you chose that approach.

Task 3 — Handle missing prices and quantity

The **Price** and **Quantity** columns have missing values. Decide whether to use the **mean** or the **median** to fill them, then apply it. Write one sentence for each column explaining your choice.

Task 4 — Handle missing delivery days

The **Delivery_Days** column is missing some values. Before filling, check whether the data has any outliers. Based on what you find, choose the right method to fill the gaps and explain your reasoning in one or two sentences.

Task 5 — Handle missing country and product

Look at the **Country** and **Product** columns carefully. These are category columns. Think about whether filling or dropping makes more sense for each one, make your decision, apply it, and justify it in one or two sentences.

Task 6 — Confirm your work

After all your cleaning, run `df.isnull().sum()` again. Copy the output into your notebook. Your result should show **zero missing values** across all columns.

Task 7 — Reflection

Write a short paragraph of five to eight sentences answering the questions below:

- Which task was the hardest decision to make, and why?
- Did you use the same method for every column?
- What would you do differently if 50% of the Price column was missing instead of just a few values?

What to Submit

A single **Jupyter Notebook** or **Python file** (.ipynb or .py) containing:

- Your code for all seven tasks
- Your written explanations as comments or markdown cells
- The reflection paragraph at the end

How You Will Be Graded

Area	What we are looking for
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Correct code	Does the code run without errors?
Right method	Did you choose the right approach for each column?
Explanation	Can you explain why you made each decision?
Reflection	Does your reflection show real thinking, not just a summary?

Remember: *There is no single correct answer for every task. What matters most is that your explanation matches your decision. A wrong method with a good reason is far better than a right method with no reason at all.*